INTRODUCTION

STUDENT INFORMATION

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REPORT OVERVIEW

This report studies five different real-world cyber incidents. It explains their background, attack methods, operational and financial effects, response measures, and practical security lessons.

KEY FOCUS

1. Ransomware, data breaches and supply-chain compromise

2. Real-world organisations, incident timelines and prevention

REPORT GUIDE

How to read this report

The report follows the same case-study structure throughout: a short incident overview, the mechanism of the attack, the damage it caused, and the security measures that could reduce similar risks.

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Reading habit

1. Identify the entry point and the system that failed.
2. Identify which controls could have prevented escalation.
3. Separate confirmed facts from reported or uncertain details.

COMMON ATTACK METHODS

Ransomware

Malicious software encrypts data or disrupts access, usually accompanied by an extortion demand.

Vulnerability exploitation

Attackers abuse a weakness in software that has not been patched or securely configured.

Supply-chain compromise

A trusted supplier, product or update mechanism becomes a route into other organisations.

Social engineering

Attackers manipulate people into disclosing credentials or approving unauthorised access.

Typical attack sequence

1. Find an entry point or trusted account.
2. Gain initial access and explore the environment.
3. Escalate access, steal information or deploy malware.
4. Disrupt operations or monetise stolen data.

ATTACK METHOD COMPARISON

Ransomware	WannaCry; Colonial Pipeline
Web vulnerability	Equifax
Software supply chain	SolarWinds
Social engineering	MGM Resorts

Important defensive controls

Security teams should prioritise timely updates, phishing-resistant multifactor authentication, least-privilege access, monitored logs, tested offline backups and an incident response plan.

Three questions to ask

1. How did the attacker gain the first foothold?
2. What let the attacker move further or cause damage?
3. Which controls could stop or detect the attack earlier?

CASE 1 | WannaCry ransomware

INCIDENT NEWS BRIEF

WANNACRY RANSOMWARE

CYBERSECURITY INCIDENT | May 2017

The WannaCry outbreak spread rapidly across vulnerable Windows computers. The UK National Health Service was among the most visible victims, with appointments and operations disrupted.

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Case snapshot

1. United Kingdom / worldwide
2. May 2017
3. Ransomware and unpatched Windows systems

CASE 1 | DETAILS

Organisation / event	WannaCry ransomware
Location	United Kingdom / worldwide
Date	May 2017
Attack type	Ransomware and unpatched Windows systems

What happened

The WannaCry outbreak spread rapidly across vulnerable Windows computers. The UK National Health Service was among the most visible victims, with appointments and operations disrupted.

How the attack worked

Attackers exploited a known Windows vulnerability to spread ransomware automatically. The malware encrypted files and demanded payment in Bitcoin.

Impact on the organisation

Hospitals and companies worldwide faced encrypted files and disrupted services. Hospitals diverted patients and staff reverted to manual processes. Organisations isolated affected networks, rebuilt devices and restored information where backups existed.

Security response

Patch operating systems, isolate infected computers, and maintain offline backups.

Main lesson

1. The outbreak showed that delayed patching can turn a known vulnerability into a global emergency.

CASE 1 | ANALYSIS

Attack timeline and stages

Four stages

1. Initial weakness: Ransomware and unpatched Windows systems.
2. Intrusion: attackers gained access or deployed malicious code.
3. Consequences: Hospitals and companies worldwide faced encrypted files and disrupted services.
4. Recovery: investigation, containment and restoration.

Prevention checklist

Patch operating systems, isolate infected computers, and maintain offline backups. Additional measures include staff awareness, event monitoring, least-privilege permissions and regular security testing.

What students should remember

The outbreak showed that delayed patching can turn a known vulnerability into a global emergency.

Discussion questions

Review

1. What was the earliest opportunity to detect this incident?
2. What security control would have reduced the damage?
3. How could the organisation restore normal operations safely?

CASE 2 | Equifax data breach

INCIDENT NEWS BRIEF

EQUIFAX DATA BREACH

CYBERSECURITY INCIDENT | 2017

Equifax disclosed a major breach involving sensitive consumer records. The incident highlighted the risk of keeping valuable personal data behind a vulnerable public-facing application.

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Case snapshot

1. United States
2. 2017
3. Exploitation of an unpatched web application vulnerability

CASE 2 | DETAILS

Organisation / event	Equifax data breach
Location	United States
Date	2017
Attack type	Exploitation of an unpatched web application vulnerability

What happened

Equifax disclosed a major breach involving sensitive consumer records. The incident highlighted the risk of keeping valuable personal data behind a vulnerable public-facing application.

How the attack worked

Attackers exploited a vulnerability in Apache Struts on an online dispute portal and accessed sensitive records over an extended period.

Impact on the organisation

Sensitive personal information of approximately 147 million people was exposed. Equifax faced investigations, legal settlements, remediation costs and widespread public concern over identity theft.

Security response

Maintain patch inventories, segment networks and monitor unusual data access.

Main lesson

1. Security updates must be tracked to completion, especially on systems connected to sensitive databases.

CASE 2 | ANALYSIS

Attack timeline and stages

Four stages

1. Initial weakness: Exploitation of an unpatched web application vulnerability.
2. Intrusion: attackers gained access or deployed malicious code.
3. Consequences: Sensitive personal information of approximately 147 million people was exposed.
4. Recovery: investigation, containment and restoration.

Prevention checklist

Maintain patch inventories, segment networks and monitor unusual data access. Additional measures include staff awareness, event monitoring, least-privilege permissions and regular security testing.

What students should remember

Security updates must be tracked to completion, especially on systems connected to sensitive databases.

Discussion questions

Review

1. What was the earliest opportunity to detect this incident?
2. What security control would have reduced the damage?
3. How could the organisation restore normal operations safely?

CASE 3 | Colonial Pipeline ransomware

INCIDENT NEWS BRIEF

COLONIAL PIPELINE RANSOMWARE

CYBERSECURITY INCIDENT | May 2021

Colonial Pipeline shut down pipeline operations after a ransomware incident affected its business network. The incident demonstrated how cyberattacks can disrupt physical supply chains.

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Case snapshot

1. United States
2. May 2021
3. Compromised VPN credentials and ransomware

CASE 3 | DETAILS

Organisation / event	Colonial Pipeline ransomware
Location	United States
Date	May 2021
Attack type	Compromised VPN credentials and ransomware

What happened

Colonial Pipeline shut down pipeline operations after a ransomware incident affected its business network. The incident demonstrated how cyberattacks can disrupt physical supply chains.

How the attack worked

Attackers used a compromised remote-access account; public reporting indicated that the account lacked multifactor authentication. DarkSide ransomware affected business systems.

Impact on the organisation

Fuel pipeline operations were temporarily shut down, contributing to supply disruption. The company halted operations as a precaution, investigated the incident and restored services over subsequent days.

Security response

Require MFA for remote access and rehearse incident response.

Main lesson

1. Even when operational equipment is not directly encrypted, a business-network incident can disrupt critical infrastructure.

CASE 3 | ANALYSIS

Attack timeline and stages

Four stages

1. Initial weakness: Compromised VPN credentials and ransomware.
2. Intrusion: attackers gained access or deployed malicious code.
3. Consequences: Fuel pipeline operations were temporarily shut down, contributing to supply disruption.
4. Recovery: investigation, containment and restoration.

Prevention checklist

Require MFA for remote access and rehearse incident response. Additional measures include staff awareness, event monitoring, least-privilege permissions and regular security testing.

What students should remember

Even when operational equipment is not directly encrypted, a business-network incident can disrupt critical infrastructure.

Discussion questions

Review

1. What was the earliest opportunity to detect this incident?
2. What security control would have reduced the damage?
3. How could the organisation restore normal operations safely?

CASE 4 | SolarWinds supply-chain attack

INCIDENT NEWS BRIEF

SOLARWINDS SUPPLY-CHAIN ATTACK

CYBERSECURITY INCIDENT | Discovered December 2020

A sophisticated campaign compromised the software supply chain of SolarWinds Orion, distributing malicious code through trusted updates.

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Case snapshot

1. United States / worldwide
2. Discovered December 2020
3. Compromised software build and malicious software updates

CASE 4 | DETAILS

Organisation / event	SolarWinds supply-chain attack
Location	United States / worldwide
Date	Discovered December 2020
Attack type	Compromised software build and malicious software updates

What happened

A sophisticated campaign compromised the software supply chain of SolarWinds Orion, distributing malicious code through trusted updates.

How the attack worked

Attackers inserted a backdoor into certain Orion software builds. Selected downstream organisations were targeted for additional access and information theft.

Impact on the organisation

Government agencies and private organisations faced espionage risks. Organisations disconnected affected systems, reviewed credentials, hunted for suspicious activity and rebuilt trusted environments.

Security response

Secure build pipelines, validate updates and limit privileged access.

Main lesson

1. Trusted vendors can become attack paths; signed software and normal update processes do not eliminate supply-chain risk.

CASE 4 | ANALYSIS

Attack timeline and stages

Four stages

1. Initial weakness: Compromised software build and malicious software updates.
2. Intrusion: attackers gained access or deployed malicious code.
3. Consequences: Government agencies and private organisations faced espionage risks.
4. Recovery: investigation, containment and restoration.

Prevention checklist

Secure build pipelines, validate updates and limit privileged access. Additional measures include staff awareness, event monitoring, least-privilege permissions and regular security testing.

What students should remember

Trusted vendors can become attack paths; signed software and normal update processes do not eliminate supply-chain risk.

Discussion questions

Review

1. What was the earliest opportunity to detect this incident?
2. What security control would have reduced the damage?
3. How could the organisation restore normal operations safely?

CASE 5 | MGM Resorts cyberattack

INCIDENT NEWS BRIEF

MGM RESORTS CYBERATTACK

CYBERSECURITY INCIDENT | September 2023

MGM Resorts experienced widespread technology disruption affecting hospitality and gaming services. The incident became a prominent example of the operational impact of identity-based attacks.

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Case snapshot

1. United States
2. September 2023
3. Social engineering and identity-system compromise

CASE 5 | DETAILS

Organisation / event	MGM Resorts cyberattack
Location	United States
Date	September 2023
Attack type	Social engineering and identity-system compromise

What happened

MGM Resorts experienced widespread technology disruption affecting hospitality and gaming services. The incident became a prominent example of the operational impact of identity-based attacks.

How the attack worked

Public accounts linked the incident to social engineering used to obtain access to internal identity and support processes. Some details of the intrusion remained disputed or limited publicly.

Impact on the organisation

Hotel and casino operations experienced major disruptions. MGM shut down or restricted systems, worked to restore services and later reported substantial financial impact.

Security response

Verify help-desk requests and strengthen identity recovery procedures.

Main lesson

1. Help-desk verification and recovery workflows deserve the same protection as passwords and multifactor authentication.

CASE 5 | ANALYSIS

Attack timeline and stages

Four stages

1. Initial weakness: Social engineering and identity-system compromise.
2. Intrusion: attackers gained access or deployed malicious code.
3. Consequences: Hotel and casino operations experienced major disruptions.
4. Recovery: investigation, containment and restoration.

Prevention checklist

Verify help-desk requests and strengthen identity recovery procedures. Additional measures include staff awareness, event monitoring, least-privilege permissions and regular security testing.

What students should remember

Help-desk verification and recovery workflows deserve the same protection as passwords and multifactor authentication.

Discussion questions

Review

1. What was the earliest opportunity to detect this incident?
2. What security control would have reduced the damage?
3. How could the organisation restore normal operations safely?

CONCLUSION & REFERENCES

What these incidents teach us

WannaCry highlights patch management; Equifax shows the danger of exposed personal records; Colonial Pipeline demonstrates operational dependence on IT systems; SolarWinds shows supplier risk; and MGM Resorts demonstrates the power of identity-focused social engineering.

Practical prevention measures

Security checklist

1. Apply security patches promptly and verify deployment.
2. Use strong MFA and carefully verify help-desk requests.
3. Restrict vendor access and segment important networks.
4. Monitor unusual activity and practise incident response.
5. Maintain offline, tested backups and recovery procedures.

Reference starting points

- CISA and Microsoft advisories on WannaCry (2017).
- US Federal Trade Commission: Equifax breach information.
- CISA / US government advisories on Colonial Pipeline and DarkSide.
- CISA Emergency Directive 21-01 on SolarWinds Orion.
- MGM Resorts public disclosures and SEC filings (2023).